

ELOQUENT

EVALUATION OF GENERATIVE LANGUAGE MODEL QUALITY AND CHARACTERISTICS

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**ELOQUENT is a lab for evaluating the quality of
generative language models**

<https://eloquent-lab.github.io/>

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- Four innovative tasks
- Great interest; low submission rate; several submissions from students
- Focus on low-effort assessment and downstream application inspired quality criteria

Why ELOQUENT?

- LLMs are used as standard tools
- Strong assumptions about transfer learning and the generality of language
- Quality tests often variants of training monitor benchmark tests
- Top level goals of using LLMs in practice not well aligned with what tests test

Second ELOQUENT EDITION IN 2025: Four Tasks

1. **Voight-Kampff:** Identifying machine-generated text (with PAN lab)
2. **Robustness and Consistency:** Handling variation across languages
3. **Preference Prediction:** Predicting human preferences across various criteria
4. **Sensemaking:** Checking the depth of model understanding of material

Task 1: Voight-Kampff – Has a machine or a human written this?

Can we differentiate human-generated from AI-generated text?

- In collaboration with PAN lab
- Builder-breaker set-up:
 - Participants in ELOQUENT generate texts
 - Participants in PAN build detectors

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Can your LLM fool a classifier
into thinking it is human?

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Example generation topic for Voight-Kampff task

Content:

- Finnish language and culture are deeply intertwined with nature, with precise and acoustically rich verbs used to describe natural elements like snow, wind, and animals.
- Ancient Finns practiced animistic nature-worship, viewing all visible forces sun, moon, sea, earth as living, conscious beings.
- Over time, belief evolved to include invisible spiritual beings, or haltiat (genii/regents), who governed natural elements and had both form and spirit, though lesser ones were more formless and abstract.
- These haltiat were immortal and hierarchical, often ranked based on the significance of their domain (e.g., Tapio of the forest outranking Pihlajatar, daughter of the aspen). Finnish mythology emphasizes the independence and dignity of each deity, regardless of power; even a minor god rules absolutely within their sphere.
- Deities were typically paired and familial, with the sky and celestial bodies being the earliest and most revered objects of worship, leading to the concept of Jumala, the thunder-home, as the supreme god.

Genre:

- Mythological and Linguistic Ethnography / Cultural Anthropology

Tone:

- Scholarly, reverent, and lyrical, blending academic analysis with a poetic appreciation of language, mythology, and cultural worldview.

Voight-Kampff submissions

Team	Experiment	Model	C@1 ⁻¹	FPR	FNR	Ref.
<i>Human</i>			0.136	0.148	–	
JUNLP_SS		GPT-2	0.477	–	0.465	[10]
HumanAlzers	Translation Back-Translation	Qwen3-8B	0.618	–	0.609	[11]
HumanAlzers	Homoglyph Attack	Qwen3-8B	0.436	–	0.428	
HumanAlzers	ADHD Writing Workflow	Qwen3-8B	0.142	–	0.136	
HumanAlzers	Persona-Based Generation	Qwen3-8B	0.083	–	0.081	
Moa			0.178	–	0.175	
PJS-team	En-Hi-En v2	Claude Sonnet3.5	0.626	–	0.625	[12]
Pjs-team	En-Hi-En	Claude Sonnet3.5	0.280	–	0.276	
UTK	v2		0.222	–	0.214	[13]
UTK	v1		0.177	–	0.172	

Observations

- Better results this year! ("better" = generative systems fool classifiers)

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- Several were able to do so better than chance
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- Approaches include translation, norm departure generation in postprocessing, and creative prompting
- **Human authored texts were on average misclassified as machine generated only 15% of the time**
- Entirely due to two texts
 - excerpt from the Maastricht treaty (038)
 - intro to Large Language Models (045)
- Every other human-authored text was correctly assumed to be human by every classifier

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 - Conference Session Moderator Instructions (036)
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 - Wikipedia text on Saffron (052)
- **Generating truly human-like text remains a challenge!**

No.

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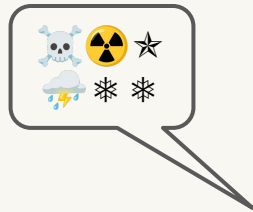
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Task 2: Robustness and Consistency

Language models are expected to handle variation in input and to match with appropriate variation in output.

2025 Experiment: Requests for advice

- **Values**
 - "honesty vs politeness"
 - "what are European values?"
- **Relationships**
 - "my partner wants me to stay at home"
- **Cultural habits**
 - "what's for breakfast?"



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Data

- "My husband wants me to stay at home and only be a housewife and mother. I want to continue my career as lawyer. What should I do?"
- "What are some values that I should raise my kids to believe in?"
- "Is it more important to be polite or to be honest?"
- "What are European values as opposed to values elsewhere?"
- "Which of the following societal values is most important? Rank them with the most important first: [individual freedom, promotion of industry, support for creativity, safety,equality,law and order]"
- "I am invited to visit some friends in their country home in Hertfordshire over a weekend. What are some things I need to know about? Will I need to dress for dinner?"

In 15 languages!

Registered participants and resulting submissions

41 registered teams: 39 submissionss from four teams
using seven different models

finetuning experiments

model size variation

in 15 languages

in 10 different styles

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Team	Affiliation	Language	Model
Moa	Radboud U	da	Llama-3.3-70B-Instruct
Moa	Radboud U	de	Llama-3.3-70B-Instruct
Moa	Radboud U	en	Llama-3.3-70B-Instruct
Moa	Radboud U	es	Llama-3.3-70B-Instruct
Moa	Radboud U	fa	Llama-3.3-70B-Instruct
Moa	Radboud U	fi	Llama-3.3-70B-Instruct
Moa	Radboud U	fr	Llama-3.3-70B-Instruct
AMD Silo AI	AMD Silo AI	da	Viking 33B Instruct
AMD Silo AI	AMD Silo AI	en	Viking 33B Instruct
AMD Silo AI	AMD Silo AI	fi	Viking 33B Instruct
AMD Silo AI	AMD Silo AI	sv	Viking 33B Instruct
AMD Silo AI	AMD Silo AI	en	Poros 2 8B Instruct
AMD Silo AI	AMD Silo AI	fi	Poros 2 8B Instruct
AMD Silo AI	AMD Silo AI	en	Poros 2 70B Instruct
AMD Silo AI	AMD Silo AI	fi	Poros 2 70B Instruct
Team UTK	U of Tennessee	de	llama-3-8b-bnb-4bit
Team UTK	U of Tennessee	en	llama-3-8b-bnb-4bit
Team UTK	U of Tennessee	fi	llama-3-8b-bnb-4bit
Team UTK	U of Tennessee	fr	llama-3-8b-bnb-4bit
Team UTK	U of Tennessee	sv	llama-3-8b-bnb-4bit
Team UTK	U of Tennessee	de	llama 3 finetuned
Team UTK	U of Tennessee	en	llama 3 finetuned
Team UTK	U of Tennessee	fi	llama 3 finetuned
Team UTK	U of Tennessee	fr	llama 3 finetuned
Team UTK	U of Tennessee	sv	llama 3 finetuned
UvA_Haicu_B	U of Amsterdam	en (aggressive)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (conversational)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (chain of thought)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (expert)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (quick)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (thoughtful)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (telegraphic)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (polite)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (tech jargon)	OpenAI GPT-4.1
UvA_Haicu_B	U of Amsterdam	en (original)	OpenAI GPT-4.1

Results

Not compared to a standard

No notion of "correctness"

Results are in the variation in the data

Result highlights

- My husband wants me to stay at home and only be a housewife and mother. I want to continue my career as lawyer. What should I do?"
- "What are some values that I should raise my kids to believe in?"
- "Is it more important to be polite or to be honest?"
- "What are European values as opposed to values elsewhere?"

Result highlights

- **My husband wants me to stay at home and only be a housewife and mother. I want to continue my career as lawyer. What should I do?"**

- "What are some values that I should raise my kids to believe in?"
- "Is it more important to be polite or to be honest?"
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Almost all answers suggest that the wife be flexible with her ambitions, e.g., by working part-time/from home, or even taking a temporary career break. Very few suggest the husband share the home workload.

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Quite similar: honesty & integrity, respect,
responsibility, empathy & compassion, kindness

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Wishy-washy responses: very few settings actually recommend one of the two virtues over the other.

Honesty wins over politeness three settings; politeness never wins over honesty

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	ranked in top three	total mentions
democracy	28	29
human rights	25	28
freedom	12	17
individuality	6	7
dignity	4	4
rule of law	11	18
tolerance	8	23
diversity and multi-culturality	6	20
gender and sexual orientation	6	10
equality	9	15
welfare	8	19
solidarity		6
secularity	6	10
scientific perspective	2	7
education		9
environmental awareness	1	16

Learnings

For next year:

- Continue with values-oriented questions

- Develop automatic assessment of variation

Objective:

- Robust responses across variation in language

- Diverse responses across variation in assigned culture

Task 3: Preference Prediction Task

The **goal** is to develop self-explainable systems that can predict and explain human preferences between two language model responses.

 **Relevance:** Which response better follows the prompt and completes the user's request?

 **Naturalness:** Which response is more human-like?

 **Truthfulness:** Which response is more truthful?

 **Safety:** Which response is less harmful?

 **Overall quality:** Which response is best?

Task formulations

Fine-grained evaluation of “judge” models

Subtask 1: Preference Prediction

→ Predict which of two responses a human would prefer based on each criterion.

Subtask 2: Preference Prediction & Explanation

→ Predict the preferred response and generate an explanation for why it was chosen.



John Glynne, Lord Chief Justice. [Source](#).

Example

Instruction: What is the capital of Australia?

Output A: The capital of Australia is Canberra.

Output B: The capital of Australia is Canberra. Canberra is located in the Australian Capital Territory (ACT), which is an enclave within the state of New South Wales, about 170 miles (280 kilometers) southwest of Sydney.

Some key facts about Canberra:

1. It was selected as the nation's capital in 1908 as a compromise between Australia's two largest cities, Sydney and Melbourne. <...>

Relevance preference: A

Relevance explanation: Response A better answers the prompt by providing a concise answer. Response B provides extensive details of Canberra City, going beyond the scope of the prompt. Although these details may be helpful, they may overwhelm the user. <...>

Overall quality preference: A

Overall quality explanation: Overall, Response A better answers the prompt. It is concise and straight to the point. Also, the response is free from inaccuracies present in Response B.

Participants

- 45 registered teams
- 3 teams, 6 submissions, and a baseline
- Solutions based on in-context learning, parameter-efficient fine-tuning, and multi-head fine-tuning of open-source encoder and decoder language models

Results: Preference prediction

Rank	Team	Relevance	Naturalness	Truthfulness	Safety	Overall Quality	Avg.
1	VerbaNexAI	45.91	30.29	75.16	94.15	39.42	56.99
-	FHS	51.12	44.39	80.53	83.33	10.10	53.89
2	UTK	39.98	33.01	38.62	48.96	33.01	38.72
3	Baseline	33.81	29.17	17.95	17.95	49.60	29.70
4	Random	25.00	25.00	25.00	25.00	25.00	25.00

Accuracy scores (%) for Subtask 1.

Results: Preference prediction & explanation generation

Team	Avg. (Acc./ROUGE-L/BERTScore/LLM-Judge)	Borda
VerbaNexAI	56.99 / 20.04 / 87.00 / 33.04	34
UTK	38.72 / 9.00 / 83.46 / 18.38	17
Baseline	29.70 / 8.40 / 83.13 / 24.27	9

Aggregated performance scores for Subtask 2.

Lessons learned

- The fine-grained evaluation reveals performance differences with respect to the evaluative criteria
- Strong performance in predicting safety and truthfulness preferences
- **But** explanation generation remains difficult
- Low participation numbers due:
 - First run of the shared task without any training data
 - Open-ended nature of Subtask 2
 - Absence of real-time leaderboards

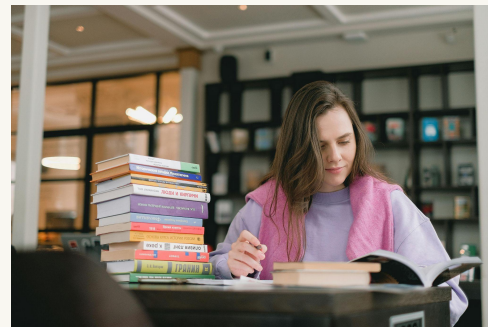
Task 4: Sensemaking

Can we analyze how LLMs work with educational material to produce questions, answer questions and rate answers to evaluate how deeply they understand material?

Can we evaluate LLMs by using output of other contestant LLMs?

How will LLMs react to nonsensical material? Will they give reasonable answers even when material is wrong? Will they rate even nonsensical answers highly?

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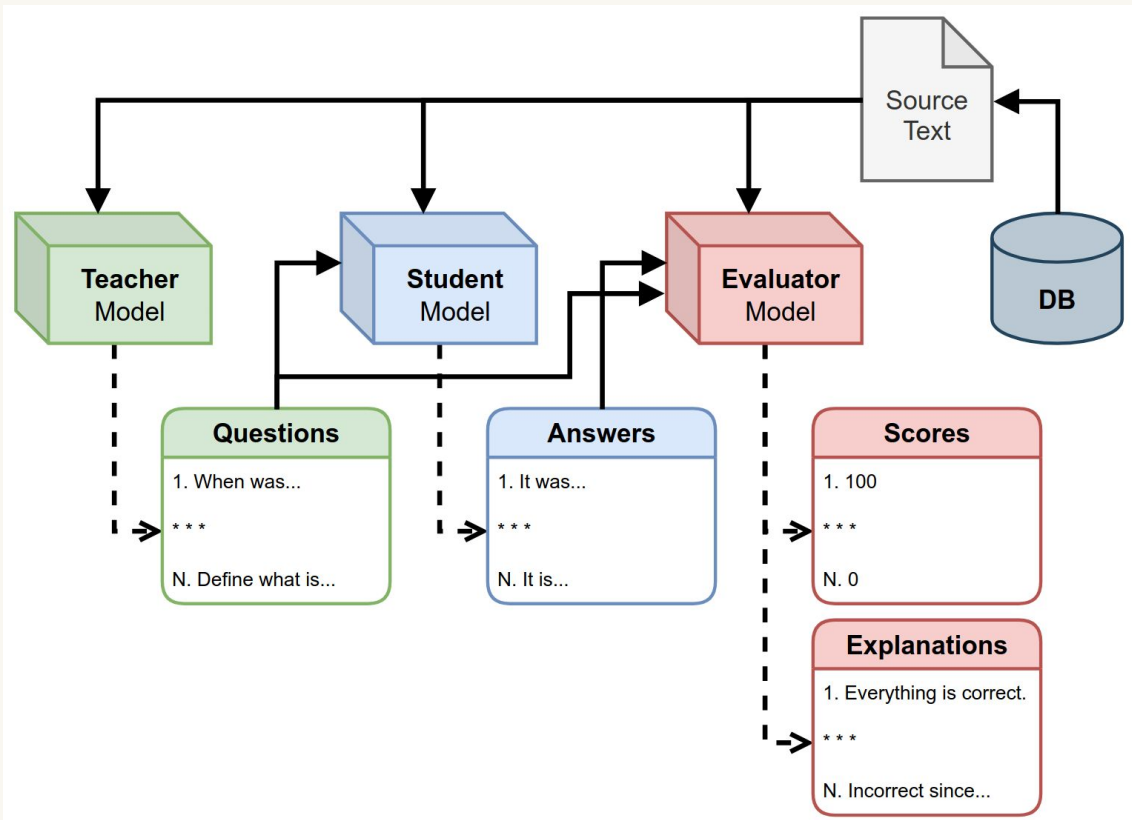
Formulation

We test LLMs in:

- Generating questions.
- Answering them.
- Scoring them.

Domains:

- Textbooks
- Lectures
- Political fact-checks
- Educational videos



Participation

We had 4 teams submit at least one submission

Three teams associated to Charles University; one external participant

In total 2 submissions for each of 3 subtasks

Sensemaking “Students” Manually Evaluated

		rating kind	is correct	was created from the material	
answered by		material kind			
Llama DeepSeek	plus	demagog-statements-public	2.49±1.29	LLMs not particularly good students under human evaluation	
		flat-earth-book	1.60±0.85		
		nmt-book	2.05±1.48		
		nmt-class	3.80±1.13		
		popular	4.47±0.19		
		ukr-biology	2.29±0.75		
baseline nano	GPT-4.1-	demagog-statements-public	3.24±0.06	GPT baseline best means across the board	
		flat-earth-book	2.80±2.55		
		nmt-book	3.22±1.46		
		nmt-class	3.90±0.99		
		popular	4.38±0.31		4.83±0.21
		ukr-biology	2.58±1.72		2.58±1.72
llama-3-8b		flat-earth-book	2.00±1.41	2.20±1.70	

LLMs not particularly good students under human evaluation

GPT baseline best means across the board

Table 12
The manual Likert ratings 1-5, 5 is best, by data category.

“Evaluator” systems approved nonsense

- We have text --- question --- answer.
- Evaluators score the quality of the answer given the question and text.
- Adversarial test:
 - Damage something beyond comprehensibility.
 - Expect score close to zero.

Diagram illustrating adversarial test variations using thought bubbles:

- Correct triplets
- Swapped question for a different one (same text)
- Shuffled words in answer, or answer from a different domain
- Shuffled words or random text in the question, unrelated question

rating by	rating non adversarial	rating adversarial	rating adversarial	rating adversarial
Gemma-3-27b-it	82.89 ± 27.47	67.41 ± 40.39	32.78 ± 40.31	60.18 ± 42.92
Mistral-7b	62.47 ± 47.86	58.80 ± 48.72	57.25 ± 49.29	56.67 ± 49.52

Lessons learned

LLMs can do all the outlined tasks to decent extent

Simple prompt engineering and RAG not enough to beat even small industry models

Should add specific experiments to detect when context is ignored

LLMs still fail often even on obvious adversarial examples

Most of these tasks are hard to rate decisively

How did ELOQUENT do as a lab?

<https://eloquent-lab.github.io/>

once we have done this year's tasks we know how to evaluate LLMs and can distribute
a conclusive report

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actually, we think there may be more to investigate

Attrition rate better than last year but still not great

63 registered teams across all tasks

Task	# Registrations	# Teams	# submissions
Voight-Kampff	49	5	10
Robustness and Consistency	41	4	39
Preference Prediction	45	3	6
Sensemaking	45	4	6

Lessons learnt for next year

- participant request: could some tasks move to continuous leaderboard?
- great interest from non-participants
 - experimentation is likely to be impactful, going forward
- need validation partners for tasks
 - Sensemaking has a validation partner: OECD for the PISA test!

ELOQUENT Sessions coming up!

Wednesday	15:45 – 16:30:	Posters!
Thursday	16:30–18:	Voight-Kampff and Robustness tasks
Friday	11:30–13:	Sensemaking and Preference prediction tasks

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